

MATH9102

INSPECTING AND PREPARING DATA FOR
PREDICTIVE MODELLING

Study Design

A careful advance plan of data collection and the analytic approach is needed to answer the question under investigation in a scientific way.

The basic elements of a study design

- Selecting an appropriate sample size for a specified level of power and level of significance
- Select appropriate measures
- Selecting methods of sampling, data collection, and analysis appropriate to the study's objectives

What do I need to describe for numerical data?

Centre

- Discuss where the middle of the data falls
- Measures of central tendency
 - mean, median and mode

Spread

- Discuss how spread out the data is
- Refers to the variability in the data
 - Range, standard deviation, IQR

Shape

- Refers to the overall shape of the distribution
- Symmetrical, uniform, skewed, or bimodal

What do I need to describe for numerical data?

Unusual Occurrences

- Outliers (value that lies away from the rest of the data)
- Gaps
- Clusters

Context

- You must write your answer
 - With reference to the context in the problem you are investigating,
 - Using **correct statistical vocabulary** adhering to referencing scheme guidelines
 - Using complete sentences.

Parametric v Non-parametric

Parametric

- Make assumptions about the population from which the sample is taken
- Shape of the population (normally distributed)

Non-parametric

- Do not make assumptions about the population and its distribution
- Tolerant set of tests which don't expect your data to anything fancy
 - Not high-powered and don't promise more than they can deliver
 - May fail to detect differences that exist
- Use for nominal or ordinal data
- Use for small samples
- Use for skewed data

Choices to be made before conducting a hypothesis test/building a model

Deciding if you have sufficient data and sufficient variability within that data

Correcting for non-response, design effect

- Weighting variable
- Be careful of scale up weighting

Missing data

- Decide what level of data is missing
- Decide what the pattern is
- Decide why it is missing
- Correct accordingly or ignore

Normality

- Inspect and test for normality
- Be aware of allowable limits
- If not normal
 - Could use non-parametric tests
 - OR
 - Apply a transformation to see if that results in a normal distribution

Linearity and homoscedasticity (scale variables)

- Inspect your scatterplot
- If assumptions are not addressed could consider transformation but also non-parametric test

Preparing your data

You may need to consider doing the following:

- Weighting your data to correct for bias, address design effects, make sample more representative of the population
- Deciding what to do about missing data
- Deciding what to do about outliers
- Recoding your variables
 - E.g. to reduce the number of categories
 - Doing so will not be objective but working with categorization at all is highly contested and highly political
- Selecting cases
 - To work only with particular sub-groups of data
- Splitting your file
 - Allows you to organise your output by category of variable you are interested in

Survey Data – Sources of Bias

Oversampling

- Some groups are purposefully sampled more frequently than others to produce a large enough sample size for analysis.

Undersampling

- Sample size is intentionally reduced to ensure equal representation of different groups.

Non-response

- When individuals who choose not to participate in a survey have different characteristics or opinions than those who participate
- This can result in under- or overestimating certain aspects of a population, leading to a skewed representation of the data.

Weighting

Use to

- Correct for any known bias that may exist in the final sample
- Scale-up frequencies so that frequencies calculated from the sample represent estimates for the population as a whole
- To address 'design effects' arising from the sampling methods used.

Example

Suppose we undertake a survey

We have selected a random sample for this purpose

Suppose we collect data on gender and marital status.

Example

Work out your actual proportions surveyed:

Demographic group	Percentage
-------------------	------------

Unmarried male	18.5%
----------------	-------

Unmarried female	18.7%
------------------	-------

Married male	21.2%
--------------	-------

Married female	20.9%
----------------	-------

Widowed man	2.3%
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Widowed woman	3.1%
---------------	------

Divorced man	4.7%
--------------	------

Divorced woman	6.6%
----------------	------

Work out your expected proportions based on population

Demographic group	Percentage
-------------------	------------

Unmarried male	20%
----------------	-----

Unmarried female	16.9%
------------------	-------

Married male	23.8%
--------------	-------

Married female	23.7%
----------------	-------

Widowed man	1.4%
-------------	------

Widowed woman	4.7%
---------------	------

Divorced man	4.1%
--------------	------

Divorced woman	5.4%
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Weighting – Correcting for Non- Response

We could conduct our analysis doing nothing to correct for this disparity, but we would be running the risk of creating biased estimates (perhaps males and females prefer different facilities)

By not correcting for bias in our sample we are giving more weight to some categories than others.

Weighting Factor

Demographic Group	Expected Distribution	Actual Distribution	Weighting Factor
Unmarried Male	20%	18.5%	1.08
Unmarried Female	16.9%	18.7%	0.90
Married Male	23.8%	21.2%	1.12
Married Female	23.7%	20.9%	1.14
Widowed Man	1.4%	2.3%	0.61
Widowed Woman	4.7%	3.1%	1.52
Divorced Man	4.1%	4.7%	0.87
Divorced Woman	5.4%	6.6%	0.82

Calculate the weights for each demographic group by dividing the expected percentage by the actual percentage

Weighting – Correcting for Non- Response

How do we use the weighting factor?

We multiply all responses for that category by the corresponding weighting factor.

We apply the weight in the calculation of all relevant statistics (e.g. weighted mean).

In R: simple multiplier applied to all variables of interest,
can include in your dataset

Weighting – Correcting for Non- Response

Weighting variables are normally used to correct for known biases in several factors.

In the youthcohort.sav (Paul Connolly) the weighting variable is used to correct for four factors related to non-response (gender, qualifications achieved, region in which they live and type of school)

Principle for calculating weighting variable remains the same.

Weighting – Correcting for design effects

Due to cluster sampling

- Used in large-scale surveys where it would be too costly to select a simple random sample

Suppose we need to conduct a survey of 10-11 yr. old pupils in school

Assuming it is possible to create a full-list of all eligible pupils in the country, then the selection of a simple random sample of 2000 pupils could mean having to interview children scattered across 800 schools throughout the country

- Lots of cost in doing interviews
- Huge amount of time in negotiating access, seeking permission etc.

Weighting — Correcting for design effects

An alternative is to use some form of cluster sampling

- Select 100 schools at random
- Survey all children in those schools
- If on average 20 pupils per school takes part this gives us our sample of 2000 pupils

But this may underestimate the amount of variation within the population as a whole

- Pupils within each cluster (each school) are likely to be more like each other than to those outside the school
- Thus, we are likely to underestimate variation unless we correct for this 'design effect'
- Consequence: increased risk of Type I error

Weighting – correcting for design effect

Correct Approach: Use multi-level modelling
(examine each cluster)

- Beyond the scope of this module

Compromise Approach: Employ a weighting variable that attempts to reduce the overall size of the sample so that standard errors are increased to account for this

In `earlychildhood.sav` and `afterschools.sav`
WTCORRECT corrects for known biases by non-response and provides a correction due to sampling.

What do I need to do?

Check if a weighting variable is included in your dataset.

If yes

- Find out what type of weighting variable it is
- Design effect – apply for all analysis
- Non-response – apply for all analysis – but check the sampling method and use caution in reporting as above

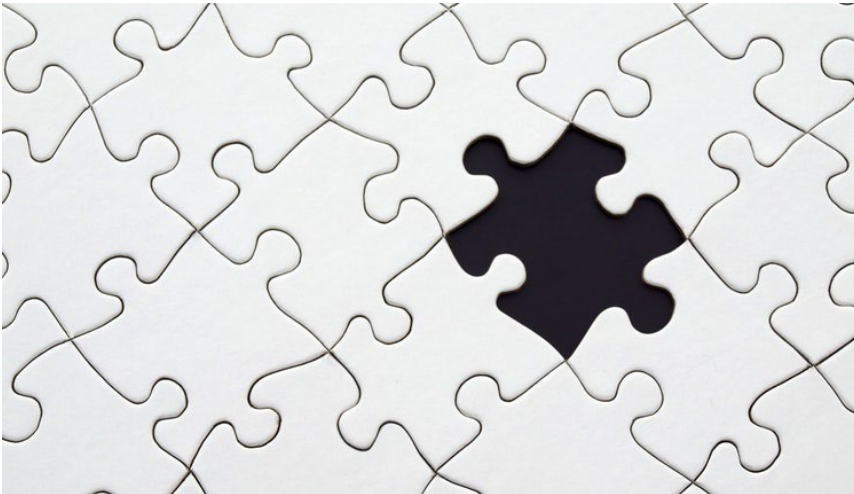
Missing Data

What is certain in quantitative research?

- Measurement error
- Missing data

Missing data can be:

- Due to preventable errors, mistakes, or lack of foresight by the researcher
- Due to problems outside the control of the researcher
- Deliberate, intended, or planned by the researcher to reduce cost or respondent burden
- Due to differential applicability of some items to subsets of respondents
- Etc.



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Why do we need to care about missing data?

Source of bias

- Introduces the possibility of making inferences on the basis of sample data that are inadvertently biased in unknown directions

Choice of treatment (e.g. deletion) can lead to loss of information and loss of statistical power through reduced sample size

Makes some common tests inappropriate or difficult to use

Why do we need to care about missing data?

The mechanism and the pattern of missing data have greater impact on results than does the amount of data missing

- The logic of statistical inference presumes that the sample is randomly drawn from the population.
- Thus, whether the missing data within a sample are random is important.
- When data are missing in a random fashion, there is no systematic difference between the available data and the missing data; they are both random subsets of the data composing the entire sample.

Evaluating Missing Data

Missing data mechanism

- Missing completely at random (MCAR)—Ignorable
- Missing at random (MAR)—Conditionally ignorable (need to identify the reason and decide)
- Missing not at random (MNAR)—Not ignorable

Evaluating Missing Data

Consider the amount of missing data

- Percent of cases with missing data
- Percent of variables having missing data
- Percent of data values that are missing

Consider the pattern of missing data

- Missing by design
- Missing data patterns

Why is it a problem?

In multivariate data a case will be excluded from the analysis if it is missing data for any variable included in the analysis.

If our sample is large, we may be able to allow cases to be excluded.

If our sample is small, we will try to use a substitution method so that we can retain enough cases to have sufficient power to detect effects.

In either case, we need to make certain that we understand the potential impact that missing data may have on our analysis.

Missing data

If missing data represent less than 5% of the total and is missing in a random pattern from a large data set, almost any procedure for handling missing values yields similar results

Tabachnik and Fidell, Using Multivariate Statistics, 6th Edition, Pearson

General Steps for Dealing with Missing Data

Identify patterns/reasons for missing and recode correctly

Understand distribution of missing data

- Consider the probability of missingness
- Are certain groups more likely to have missing values?
- Example: Respondents in service occupations less likely to report income
- Are certain responses more likely to be missing?
- Example: Respondents with high income less likely to report income

General Steps for Dealing with Missing Data

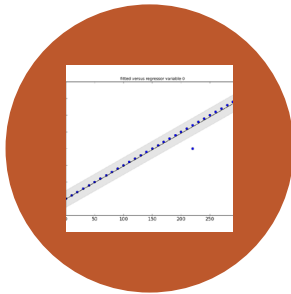
Decide on best method of analysis

- Use what you know about
- Why data is missing
 - Distribution of missing data

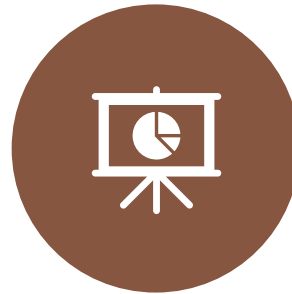
Decide on the best analysis strategy to yield the least biased estimates

- Deletion Methods
 - Listwise deletion, pairwise deletion
- Single Imputation Methods
 - Mean/mode substitution, dummy variable method, single regression
- Model-Based Methods
 - Maximum Likelihood, Multiple imputation

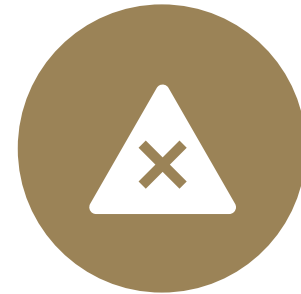
Outliers



CASES THAT HAVE DATA VALUES THAT ARE VERY DIFFERENT FROM THE DATA VALUES FOR THE MAJORITY OF CASES IN THE DATA SET.



IMPORTANT BECAUSE THEY CAN CHANGE THE RESULTS OF OUR DATA ANALYSIS.



WHETHER WE INCLUDE OR EXCLUDE OUTLIERS FROM A DATA ANALYSIS DEPENDS ON THE REASON WHY THE CASE IS AN OUTLIER AND THE PURPOSE OF THE ANALYSIS.

Univariate and Multivariate Outliers

Univariate outliers

- Cases that have an unusual value for a single variable

Multivariate outliers

- Cases that have an unusual combination of values for several of the variables.
- The value for any of the individual variables may not be a univariate outlier, but, in combination with other variables, is a case that occurs very rarely.

Outliers

Reasons for outliers

- Data entry error
- Failure to specify a particular value for missing data
- Outlier not a true member of population of interest
- Outlier is a true member of population of interest with an extreme score

What to do?

- Transform to standardized variables
 - Look at histogram
 - Sometimes transforming data can “pull in” the outlier
- Censoring outliers
 - May need to delete case/s and run with and without outlier

Standard Scores Detect Univariate Outliers

One way to identify univariate outliers is to convert all values for a variable to standard scores.

If the sample size is small (80 or fewer cases)

- a case is an outlier if its standard score is ± 2.5 or beyond.

If the sample size is larger than 80 cases

- a case is an outlier if its standard score is ± 3.29 or beyond

This method applies to interval/ratio level variables, and to ordinal level variables that are treated as metric.

It does not apply to nominal level variables.

Transformations to Improve Normality

Many statistical methods require that the numeric variables you are working with have an approximately normal distribution.

Reality is that this is often not the case.

One of the most common departures from normality is skewness.

There are many different types of transformation available.

Tools for Assessing Normality

According to some researchers, sometimes violations of normality are not problematic for running parametric tests.

Our Heuristic Approach is based on this premise:

- Inspect Histogram and Boxplot
- Inspect Normal Quantile Plot
(also called Normal Probability Plot)
- Generate and Inspect Recognized heuristics
 - Standardized skewness and kurtosis
 - Percentage of standardized scores falling within ranges (see previous slides and previous lectures for bounds)

Tools for Assessing Normality

Goodness of Fit Tests

- Shapiro-Wilk Test (small dataset)
- Kolmogorov-Smirnov Test (larger dataset)
- These are unreliable if your dataset is large
- Anderson-Darling Test: extension of the Kolmogorov-Smirnov test and gives more weight to the tails of the distribution
- D'Agostino-Pearson Test: combines skewness and kurtosis to assess normality. It's more suitable for larger datasets, typically above 50 observations, and provides a good balance of sensitivity and specificity.

Transforming Data

When a variable is not normally distributed (a distributional requirement for many different analyses), we can create a transformed variable and test it for normality.

Transformation:

- Perform a mathematical operation on each of the scores in a set of data, and thereby converting the data into a new set of scores which are then employed to analyze the results of an experiment.
- If the transformed variable is normally distributed, we can substitute it in our analysis.

Transforming Data

Transformations are obtained by computing a new variable.

- R, either transform and save variable to dataset or apply transformation within whatever statistical formula

There may be data values which are not mathematically permissible.

- For example, the log of zero is not defined mathematically, division by zero is not permitted, and the square root of a negative number results in an “imaginary” value.
- You need to decide what to do about this.

Transformations

Logarithmic transforms

- “Pull in” distribution toward a more normal curve.
- Can be used for right (positive) skew, positive kurtosis, unequal variance, lack of linearity ($\log_{10}$).
- Reduces skewness by compressing high values more than low values.

Square root transformation

- Will bring larger scores closer to the centre
- Can be used for positive skew, positive kurtosis, unequal variance, lack of linearity.
- Especially for count data or when values are close to zero.
- Reduces the impact of extreme values while still preserving the data's original scale.
- Can also use **cube root**

Transformations

Reciprocal transformation ($1/x$)

- Reduces impact of large scores
- Can be used for positive skew, positive kurtosis, unequal variance, lack of linearity
- Often used when there are large outliers.

Combination

- **Reverse score** transformation and then apply one of the above transformations
- For negatively skewed/kurtotic data

Transformations

Box-cox transformation

- Flexible family of power transformations that can accommodate various types of skewness by adjusting the power parameter λ lambda.
- $\lambda = 1$: No transformation is applied; the data remains unchanged.
- $\lambda = 0$: The transformation becomes a natural log, $\ln(y)$, which is effective for reducing right skewness in positively skewed data.
- $0 < \lambda < 1$: A power transformation that can moderately reduce right skewness while still compressing larger values
- $\lambda < 0$: Negative values of λ apply reciprocal transformations that can be effective for highly skewed data with large outliers
- $\lambda > 1$: A stronger power transformation that might be useful for left-skewed data

Transformations

Two quick videos

Short and to the point:

https://www.youtube.com/watch?v=5571wc0iWCI&ab_channel=Udacity

More involved:

https://www.youtube.com/watch?v=kV9_qFR3ORg&ab_channel=PatrickManapat

Transforming Data

If transformation does not bring data to a normal distribution, the investigators might well choose a **nonparametric** procedure that does not make any assumptions about the shape of the distribution.

Good Information Source

Andy Field's The Beast of Bias

<https://www.discoveringstatistics.com/repository/exploringdata.pdf>